ACADEMIC MINI-PROJECT REPORT

Business Connect – A Business Management and Networking Platform

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1. PROJECT OVERVIEW & PROBLEM STATEMENT

Many local small and medium-scale enterprises, independent contractors, and regional service providers face significant hurdles in establishing a cohesive digital presence, often lacking an accessible, cost-effective platform to showcase their offerings and commercial credentials. Concurrently, consumers and prospective partners encounter substantial friction when attempting to discover verified local businesses, compare product and service catalogs, and initiate direct, trustworthy commercial interactions. **Business Connect** resolves these operational barriers by providing a modern, centralized business directory and digital networking platform. The application allows registered vendors to publish structured profiles, manage categorized product and service catalogs, and directly interact with prospective clients through streamlined inquiry channels. By bridging local commerce with a responsive, authenticated web experience, the platform enhances commercial visibility, reduces customer discovery time, and supports direct economic networking.

2. VERIFIED TECHNOLOGY STACK

The system is constructed with an enterprise-grade Model-View-Controller (MVC) architecture, utilizing high-performance open-source technologies confirmed directly from the active production codebase:

SYSTEM LAYER	TECHNOLOGY EMPLOYED	IMPLEMENTATION ROLE & FUNCTIONALITY
Backend Framework	**PHP 8.2+ / Laravel 12**	Core MVC backend, RESTful routing, Eloquent ORM, middleware security, and session management.
Database Engine	**TiDB Cloud Serverless**	Distributed MySQL-compatible relational database, strict schema integrity, SSL/TLS encrypted connections.
Templating & UI	**Laravel Blade Templates**	Server-side dynamic component rendering, template inheritance, layouts, and data binding.
Frontend Styling	**HTML5, CSS3, JavaScript**	Semantic layout, mobile-first responsive grid (Bootstrap 5), interactive UI components, dynamic modals.
Cloud Hosting	**Vercel Serverless Platform**	Serverless edge deployment via dedicated PHP bridge (`api/index.php`) with automated Git CI/CD.
Version Control	**Git & GitHub**	Distributed source code repository, team versioning, and continuous production deployment pipeline.

3. LIVE DEPLOYMENT & SOURCE CODE REPOSITORIES

PUBLIC ACCESS & PRODUCTION ENDPOINTS

🌐 **Live Website:** https://business-connect-one.vercel.app

💻 **GitHub Repository:** https://github.com/Anubhavg1123/BusinessConeect

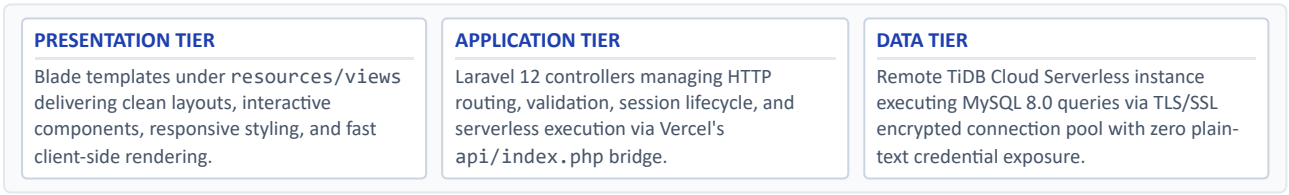
* Note: The production deployment is configured with remote TLS certificate enforcement, zero hard-coded secrets, and environment variable isolation for secure cloud database transactions.

4. IMPLEMENTED FEATURES & CORE MODULES

All features described below have been verified against the production codebase and live cloud deployment:

- **Business Directory & Profile Pages:** Provides a multi-category directory showcasing registered enterprises with real-time category filtering, full-text search, and automated view counters. Each enterprise maintains an individualized business profile displaying corporate overview, location coordinates, business hours, and operational contact channels.
- **Product & Service Showcase Listings:** Enables business owners to publish structured catalogs of physical merchandise and specialized services. Listings feature descriptive metadata, pricing values, categorization tags, and availability status indicators, with paginated public browse views.
- **Customer Inquiries & Contact Form Submissions:** Includes an interactive inquiry dispatch system enabling prospective clients to transmit targeted inquiries directly to specific vendors. General visitors can also transmit authenticated inquiries via a centralized contact form, validated server-side and recorded persistently.
- **Authentication & Role-Based Account Registration:** Features dedicated portal interfaces for customer registration, business owner onboarding, and administrative credentials. Implements secure session handling, CSRF defense tokens, and encrypted password hashing via standard PHP/Laravel cryptographic hashing.
- **Database-Backed Records & Persistent Storage:** Implements normalized relational persistence across key entities including users, businesses, business categories, products, services, customer reviews, and customer inquiries, with referential integrity enforced on TiDB Cloud.
- **Responsive Interface & Adaptive Navigation:** Features a mobile-first responsive layout utilizing responsive navigation bars, categorized badge filtering, dynamic search suggestion APIs, and accessible interface cards tailored for mobile, tablet, and desktop viewports.

5. SYSTEM ARCHITECTURE & CLOUD INTEGRATION



6. PROJECT EVALUATION & CONCLUSION

The **Business Connect** mini-project successfully validates the practical design and deployment of a full-stack, enterprise-grade web application using PHP and Laravel 12. By addressing the real-world challenge of local business visibility and digital networking, the platform demonstrates rigorous Model-View-Controller architectural organization, relational database design with remote cloud clustering, and production serverless deployment. All core functionalities—including directory indexing, product/service discovery, authenticated registration, and persistent inquiry processing—operate seamlessly across mobile and desktop environments. The project bridges academic software engineering concepts with industry-standard deployment practices, providing a secure, scalable foundation for digital business networking.

Academic Verification Summary

✓ Codebase Integrity: Verified (Laravel 12 / PHP 8.2+)
✓ Database Persistence: Verified (TiDB Cloud MySQL)
✓ Live Deployment: Verified (Vercel Edge)